

# G7 Term 1

## Term Break Homework

Name: \_\_\_\_\_

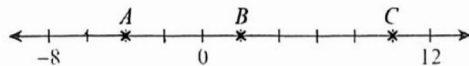
Class: \_\_\_\_\_ Date: \_\_\_\_\_

Marks: \_\_\_\_\_ / 45

1. Let  $1\blacktriangle\blacktriangle6$  be a 4-digit number. If  $1\blacktriangle\blacktriangle6$  is divisible by 4, find the possible values of  $\blacktriangle$ .  
(2 marks)

2. Along a straight road from a bus stop to a library, a lamp post and a garbage bin stand together 50 m away from the bus stop. Then, there are a lamp post at every 150 m and a garbage bin at every 500 m on the road. If a lamp post and a garbage bin stand together again outside the library, find the distance (in m) between the bus stop and the library. (3 marks)

3. Consider the following number line.



- (a) Write down the numbers represented by  $A$ ,  $B$  and  $C$ . (3 marks)
- (b) Find the value of  $\frac{A-C}{-2B}$ . (2 marks)

4. The following table shows the positions of 5 hikers at a hiking trail now, where '+1 m' represents 1 m above the starting point of the trail.

| Hiker        | A   | B   | C   | D   | E   |
|--------------|-----|-----|-----|-----|-----|
| Position (m) | -20 | +45 | +10 | -35 | -25 |

- (a) Arrange their positions in ascending order. (1 mark)
- (b) If the starting point of the trail is 210 m above the sea level, what is the average height of these 5 hikers above the sea level? (2 marks)

5. The general term  $a_n$  of a sequence is  $7n-1$ .

(a) Find  $a_{20}$ .

(2 marks)

(b) If  $a_r = 2a_8 - a_{10}$ , find the value of  $r$ .

(2 marks)

6. Solve the equation  $2(3x-1)-3(6+x)=8x-10$ .

(3 marks)

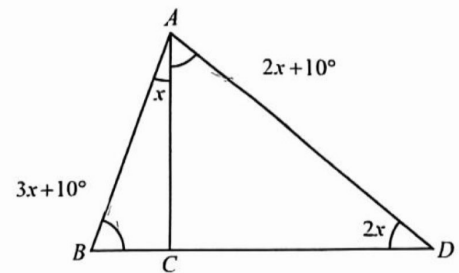
7. On a table, the number of brand B light bulbs is more than that of brand A light bulbs by 12. Every 6 brand A light bulbs or every 4 brand B light bulbs are packed in a box. If all light bulbs fill up 18 boxes, find the total number of light bulbs on the table.

(4 marks)

8. In a box, there are  $x$  blue marbles, some red marbles and some green marbles. The number of red marbles is more than the number of blue marbles by  $x + 4$ , and it is also 2 times the number of green marbles. It is given that each marble weighs  $y$  grams.
- (a) Let  $W$  grams be the total weight of the marbles in the box. Find the formula for  $W$ . (3 marks)
- (b) Find the value of  $x$  if  $y = 4.5$  and  $W = 387$ . (2 marks)

9. In the figure,  $BCD$  is a straight line.

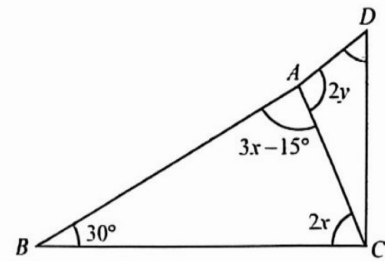
- (a) Is  $\triangle ABD$  an acute-angled triangle? Explain your answer. (4 marks)
- (b) Is  $\triangle ACD$  a right-angled triangle? Explain your answer. (3 marks)



10. The figure shows a quadrilateral  $ABCD$ . It is given that  $\angle BCD$  is a right angle.

(a) Find  $x$  and  $y$ . (6 marks)

(b) Is  $ABCD$  a convex polygon? Explain your answer. (3 marks)



END OF PAPER